Asynchronous Two-Tier Anomaly Triage for Radar Telemetry Using Masked Diffusion Under RF Blackouts

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Abstract*—Detecting anomalies in radar satellite telemetry on edge processors is heavily constrained by Size, Weight, and Power limitations and the physical reality of discontinuous RF data transmissions. Standard deep learning anomaly detectors rely on continuous sliding-window evaluations, creating severe computational overhead and forcing numerical interpolation across routine communication gaps, such as 149-second RF blackouts. This induces artificial high-frequency transients and false alarms. In this paper, we propose an asynchronous two-tier anomaly triage and heuristic diagnostic cascade tailored for resource-constrained radar telemetry. Layer 1 operates as a near-real-time edge triage reflex, utilizing a per-channel tail-calibrated Isolation Forest augmented with burst-isolated temporal features and closed-form Ridge regression residuals to flag suspicious telemetry segments. Layer 2 acts as a high-specificity verification engine, employing a lightweight Observation-Masked Frequency-Guided Diffusion model that restricts noise estimation and Nadaraya-Watson frequency prior extraction strictly to observed sequence indices, eliminating interpolation artifacts. Evaluated across 54 telemetry channels from the NASA Soil Moisture Active Passive (SMAP) radar spacecraft under simulated RF blackouts, the proposed cascade bypasses 97.4 percent of dense stride-1 sliding-window diffusion evaluations (a 38.1-fold cascade speedup). By rejecting 52.9 percent of spurious Layer 1 candidate windows, Layer 2 elevates step-level precision from 33.96 percent to 45.24 percent, increases Point-Adjusted Precision from 50.99 percent to 64.36 percent, and achieves an end-to-end Point-Adjusted F1-score of 0.7004 while heuristically classifying the anomaly type (Point versus Contextual) with 72.7 percent accuracy.

***Index Terms*—Anomaly Detection, Radar Telemetry, Edge Triage, Diffusion Models, Irregular Time Series, RF Blackouts**

I. Introduction

While modern radar payloads, such as the L-band Synthetic Aperture Radar (SAR) aboard the NASA SMAP satellite, generate high-resolution active sensing data, they are highly vulnerable to platform-level anomalies. In July 2015, the SMAP radar suffered a permanent hardware failure [1]. Safeguarding active radar instruments requires rigorous, low-latency monitoring of the spacecraft's underlying housekeeping telemetry (e.g., power buses, thermal loops, and actuator commands) even when the primary RF downlink experiences severe line-of-sight dropouts. This paper does not analyze radar return signals; rather, it focuses on the mission-critical task of edge-based anomaly triage for the radar's host platform, ensuring survival during discontinuous telemetry reception.

Deploying state-of-the-art multivariate time-series anomaly detection (MTSAD) faces two fundamental systems barriers: strict Size, Weight, and Power (SWaP) limits on edge processing hardware, and the gapped nature of RF telemetry. Recent deep generative architectures—including Long Short-Term Memory (LSTM) predictors [1], Variational Autoencoders (VAEs) [3], Attention-based Transformers [4], and Denoising Diffusion Probabilistic Models (DDPMs) [2], [5], [6]—achieve strong reconstruction accuracy on benchmark datasets but assume two conditions rarely met in operational radar flight data:

1) **Dense Sliding-Window Compute:** Standard deep models evaluate every incoming timestamp via a stride-1 sliding window. Running hundreds of iterative denoising steps at every telemetry tick rapidly saturates compute budgets, especially when over 85% to 95% of nominal mission telemetry is entirely quiescent.
2) **Uniform Temporal Grids:** Spectral and frequency-guided diffusion models rely on continuous Fourier or Hilbert-Blaschke decompositions [2], [7] that require uniformly spaced samples ($\Delta t = 1$). Bridging a 149-second RF blackout via linear or cubic spline interpolation fabricates synthetic trajectories whose sharp boundary derivatives trigger bursts of false-positive alarms.

To overcome both bottlenecks without fabricating data across communication blackouts, we propose an **Asynchronous Two-Tier Anomaly Triage Cascade** combining an edge tree reflex (**Layer 1**) with an observation-masked frequency-guided diffusion verifier (**Layer 2**). Rather than forcing Layer 1 to act as a standalone classifier, we formulate Layer 1 as a high-sensitivity hardware trigger (PA-Recall $= 85.39\%$) operating on a micro-batch buffer that routes only compact, suspicious candidate windows ($W = 32$) to Layer 2. Layer 2 then applies a binary observation mask $\mathbf{M} \in [0,1]^{W \times D}$ over the true irregular sequences, decomposing

the observed signal into low-frequency trends ($\mathbf{P}_{\text{low}}$) and high-frequency transients ($\mathbf{P}_{\text{high}}$) via sequence-masked kernel smoothing to filter out false alarms and heuristically diagnose the fault class.

The primary contributions of this work are summarized as follows:

- **Burst-Isolated Edge Triage (Layer 1):** We design a per-channel Isolation Forest operating on burst-grouped temporal derivatives and a closed-form Ridge regression command-to-sensor residual. By computing dynamic features strictly within contiguous telemetry bursts and calibrating thresholds on the extreme tail of nominal training data, Layer 1 eliminates blackout-boundary leakage.
- **Observation-Masked Frequency-Guided Diffusion (Layer 2):** We introduce a lightweight conditional diffusion verifier (*Lite-PhysDiff*) that replaces continuous Fourier basis searches with a masked Nadaraya-Watson frequency decomposition and masked noise-prediction loss, preventing missing blackout intervals from corrupting reconstruction scores.
- **Heuristic Anomaly Taxonomy:** By comparing high-frequency primary-sensor impulse ratios against cross-channel auxiliary command reconstruction error, Layer 2 heuristically classifies confirmed anomalies into *Point* versus *Contextual* faults with 72.7% accuracy.
- **Empirical Validation on Gapped Radar Telemetry:** Across 54 channels of the NASA SMAP radar dataset subjected to 90% blackout decimation (149-second gaps), our cascade reduces diffusion window invocations by 97.4% ($38.1\times$ speedup over dense sliding windows), rejects 52.9% of Layer 1 false-alarm windows, and improves Point-Adjusted F1 from 0.6385 to 0.7004.

II. Related Work

A. Radar and Spacecraft Telemetry Anomaly Detection

Hundman et al. [1] introduced the NASA Soil Moisture Active Passive (SMAP) benchmark alongside Telemanom, an LSTM-based forecasting framework with dynamic error thresholding. Subsequent architectures incorporated graph neural networks [8] and discrepancy-based attention mechanisms [4] to capture cross-sensor correlations. While effective on continuous ground archives, autoregressive and dense-attention models incur high latency and degrade when streams suffer from asynchronous RF dropouts.

B. Diffusion Models and Irregular Time Series

Denoising Diffusion Probabilistic Models (DDPMs) [5] have recently been adapted for time-series imputation and anomaly detection, notably in TimeGrad [6] and CSDI [9]. Physics-guided diffusion approaches, such as PhysDiff [2], further integrate frequency-domain priors [7] to guide denoising trajectories. However, classical spectral decompositions require uniform sampling; when severe RF blackouts occur, missing segments destabilize complex phase estimation. Conversely, lightweight tree ensembles such as Isolation Forest [10] offer fast inference but lack temporal receptivity. Our work bridges this gap by coupling a physics-augmented Isolation Forest trigger with an observation-masked diffusion verifier.

III. Proposed Two-Tier Triage Architecture

A. Problem Formulation and Asynchronous Grid

Let a subsystem channel consist of D telemetry variables ($D = 25$), where dimension 0 ($x_{t,0} \in \mathbb{R}$) denotes the continuous primary telemetry sensor and dimensions $1, \ldots, D-1$ ($\mathbf{u}_t \in \{0,1\}^{D-1}$) represent discrete command and state flags. Under irregular RF blackouts, observations arrive at strictly increasing integer timestamps $\{t_1, t_2, \ldots, t_N\}$ with inter-arrival intervals $\Delta t_i = t_i - t_{i-1} \geq 1$. Whenever $\Delta t_i > 1$, a communication blackout of duration $\Delta t_i - 1$ steps has occurred. We partition the irregular sequence into contiguous telemetry bursts indexed by:

$$b_i = \sum_{k=1}^{i} \mathbb{I}(\Delta t_k > 1), \tag{1}$$

where $\mathbb{I}(\cdot)$ is the indicator function. All Layer 1 temporal operators are strictly confined within each burst group b_i.

B. Layer 1: Near-Real-Time Burst-Isolated Edge Triage

To maintain signal stability at blackout boundaries while preserving low latency, Layer 1 operates with a minor 5-step micro-batch buffer. We extract an 8-dimensional feature vector $\mathbf{f}_i \in \mathbb{R}^8$ at each step i:

1) *Raw Primary Amplitude:* $f_{i,0} = x_{i,0}$.
2) *Nominal Baseline Z-Deviation:* Using nominal training mean μ_0 and standard deviation σ_0 (floored at 10^{-3}), $f_{i,1} = |x_{i,0} - \mu_0|/\sigma_0$.
3) *Burst-Isolated Velocity:* $f_{i,2} = |x_{i,0} - x_{i-1,0}|$ for $b_i = b_{i-1}$, with the initial sample of each burst initialized to zero to prevent artificial spikes.
4) *Local Rolling Mean Deviation:* $f_{i,3} = |x_{i,0} - \bar{x}^{(b)}_{i,5}|$, where $\bar{x}^{(b)}_{i,5}$ is the 5-step causal mean within burst b_i.
5) *Local Rolling Volatility:* $f_{i,4} = \text{std}^{(b)}_5(x_{i,0})$.
6) *Sustained Oscillation Range:* $f_{i,5} = \max^{(b)}_{10}(x_{i,0}) - \min^{(b)}_{10}(x_{i,0})$.
7) *Ridge Command-Mismatch Residual:* We fit a closed-form ℓ_2-regularized Ridge regression ($\alpha = 1.0$) on nominal training data using strictly prior regressors $\mathbf{z}_i = [x_{i-1,0}, \bar{x}^{(b)}_{i-1,5}, \mathbf{u}_i^\top]^\top$:
$$f_{i,6} = \left| x_{i,0} - \left(\mathbf{w}^\top \mathbf{z}_i + c \right) \right|. \tag{2}$$
8) *Auxiliary Command Novelty:* Given per-command training means $\boldsymbol{\mu}_{\text{aux}}$ and standard deviations $\boldsymbol{\sigma}_{\text{aux}}$:
$$f_{i,7} = \max_{1 \leq d < D} \frac{|u_{i,d} - \mu_{\text{aux},d}|}{\sigma_{\text{aux},d} + 10^{-2}}. \tag{3}$$

For each channel, an Isolation Forest (150 trees) is trained. We standardize the negative path-length score r_i against the

training score distribution (μ_r, σ_r) and combine it with a robust tail deviation $z_{\text{emp},i} = (r_i - \text{median}(r_{\text{train}}))/\text{MAD}(r_{\text{train}})$:

$$s_i = \frac{r_i - \mu_r}{\sigma_r + 10^{-6}} + 0.35 \cdot \text{clip}(z_{\text{emp},i} - 3.0,\ 0,\ 15)\,. \tag{4}$$

Dual hysteresis thresholds are calibrated on the nominal training tail:

$$\begin{aligned} \tau_{\text{high}} &= \max(\text{P}_{98.5}(s_{\text{train}}),\ 2.2)\,, \\ \tau_{\text{low}} &= \max(\text{P}_{92.0}(s_{\text{train}}),\ 1.3)\,. \end{aligned} \tag{5}$$

If any step within burst b_i exceeds τ_{high}, adjacent steps satisfying $\tilde{s}_i > \tau_{\text{low}}$ are coalesced into candidate windows of length $W = 32$ and routed to Layer 2.

C. Layer 2: Observation-Masked Frequency Diffusion

When Layer 1 routes a candidate window $\mathbf{X}$, Layer 2 constructs an observation mask $\mathbf{M} \in [0,1]^{W \times D}$, where $M_{\tau,d} = 1.0$ for observed steps and 0.0 for tail padding.

1) Sequence-Masked Frequency Decomposition: Rather than fabricating data across the gap via interpolation, the kernel operates exclusively on the topology of the received packet sequence (observation index τ), isolating high-frequency spikes from the nominal downlinked trend without bridging the temporal void. We define the Gaussian smoothing kernel $K(\tau, j) = \exp\big(-(\tau - j)^2/(2\sigma_k^2)\big)$ with bandwidth $\sigma_k = 2.5$ and extract the masked trend prior $\mathbf{P}_{\text{low}}$:

$$\mathbf{P}_{\text{low}}(\tau) = \mathbf{M}(\tau) \odot \frac{\sum_{j=1}^{W} K(\tau, j)\, \mathbf{M}(j) \odot \mathbf{X}(j)}{\sum_{j=1}^{W} K(\tau, j)\, \mathbf{M}(j) + \epsilon}, \tag{6}$$

The high-frequency transient prior is obtained as the masked residual:

$$\mathbf{P}_{\text{high}}(\tau) = \mathbf{M}(\tau) \odot (\mathbf{X}(\tau) - \mathbf{P}_{\text{low}}(\tau))\,. \tag{7}$$

2) Frequency-Routing Conditional Denoiser: We formulate a compact conditional diffusion network ($d_h = 64$, $T = 20$). At step t, the forward process corrupts observed entries via $\mathbf{X}_t = \mathbf{M} \odot (\sqrt{\bar{\alpha}_t}\mathbf{X}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon})$. Inside the denoiser, latent projections $\mathbf{H}_x, \mathbf{H}_l, \mathbf{H}_h$ are dynamically weighted via a softmax gate [2]:

$$[g_{\text{low}}, g_{\text{high}}] = \text{softmax}(\mathbf{W}_g[\mathbf{H}_x \,\|\, \mathbf{H}_l \,\|\, \mathbf{H}_h] + \mathbf{b}_g)\,, \tag{8}$$

$$\mathbf{H}_{\text{fused}} = \mathbf{H}_x + g_{\text{low}} \odot \mathbf{H}_l + g_{\text{high}} \odot \mathbf{H}_h + \mathbf{e}_{\text{step}}(t), \tag{9}$$

predicting the noise $\hat{\boldsymbol{\epsilon}}_\theta$. The network is trained with a masked noise loss and a one-step frequency-consistency regularizer:

$$\mathcal{L}_{\text{total}} = \frac{\|\mathbf{M} \odot (\hat{\boldsymbol{\epsilon}}_\theta - \boldsymbol{\epsilon})\|_F^2}{\|\mathbf{M}\|_1 + \epsilon} + \frac{\lambda_{\text{phys}}}{\|\mathbf{M}\|_1 + \epsilon} \sum_{c \in \{\text{low, high}\}} \left\| \mathbf{M} \odot (\hat{\mathbf{P}}_c - \mathbf{P}_c) \right\|_F^2, \tag{10}$$

During inference, we perform seeded deterministic analytical reconstruction averaged across noise levels $t \in \{4, 10, 16\}$.

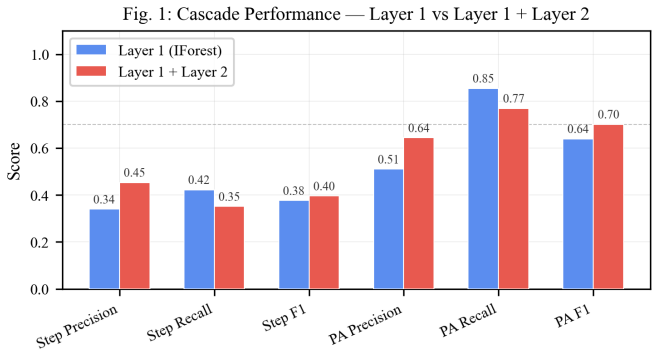


Fig. 1. Comparison of detection performance between Layer 1 (edge triage) and the Layer 1 + Layer 2 cascade. Layer 2 purges false-positive bursts.

TABLE I
ANOMALY DETECTION ON GAPPED NASA SMAP (54 CHANNELS)

Evaluation Metric	Layer 1 Alone	Layer 1 + Layer 2
Step-Level Precision	0.3396	**0.4524** (+11.28%)
Step-Level Recall	**0.4221**	0.3513
Step-Level F1-Score	0.3764	**0.3955** (+1.91%)
Point-Adjusted Precision	0.5099	**0.6436** (+13.37%)
Point-Adjusted Recall	**0.8539**	0.7682
Point-Adjusted F1-Score	0.6385	**0.7004** (+6.19%)
Stride-1 Window Computes	41,298	**1,085**
Candidate Windows Rejected	0 / 1,085 (0.0%)	**574 / 1,085 (52.9%)**
Diffusion Evals Saved	—	**97.4% (38.1× Speedup)**
Taxonomy Accuracy (Detected)	N/A	**72.7% (16 / 22)**

D. False-Alarm Filtering and Heuristic Anomaly Taxonomy

Layer 2 computes primary high-frequency error $e_{\text{HF}}(\tau)$, low-frequency error $e_{\text{LF}}(\tau)$, combined score $e_{p0}(\tau)$, and auxiliary error $e_{\text{aux}}(\tau)$ over observed steps. To suppress false alarms, a window is rejected if: (i) the primary sensor is flat in training ($\sigma_{0,\text{train}} < 0.015$) with no excursion; (ii) window variance is static without a mean shift; or (iii) errors fall below calibrated validation thresholds without an amplitude deviation.

For confirmed anomalies, Layer 2 heuristically diagnoses the anomaly type by evaluating the excursion δ_{oob}, the high-frequency peak-to-median ratio κ_{HF}, and the auxiliary coupling ratio ρ_{aux}. Anomalies exhibiting direct range excursions ($\delta_{\text{oob}} > 0.02$), sharp uncoupled impulses ($\kappa_{\text{HF}} > 4.5$ and $\rho_{\text{aux}} < 0.85$), or low auxiliary dimensionality are classified as **Point** anomalies, whereas multi-channel command-coupled deviations are classified as **Contextual** anomalies.

IV. Experimental Setup and Results

A. Dataset and RF Blackout Protocol

We evaluate the architecture on the NASA SMAP satellite benchmark [1], an Earth-observing mission equipped with an L-band radar. The original SMAP benchmark contains 55 channels with variable encoded command dimensions and

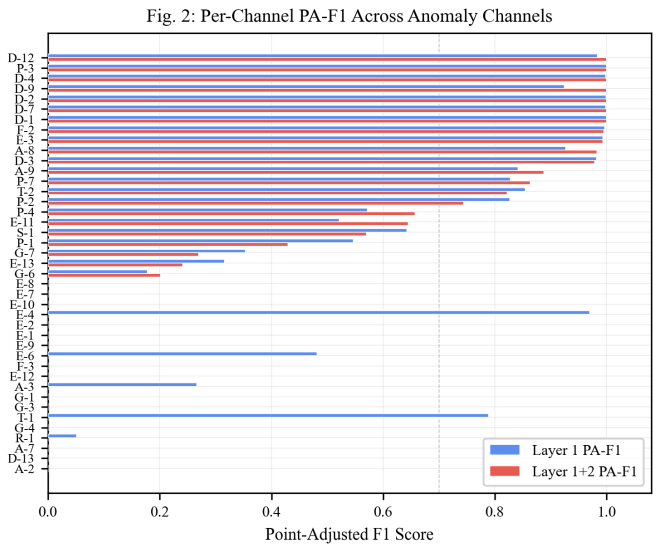


Fig. 2. Per-channel Point-Adjusted F1 scores on anomalous SMAP channels.

anonymized timestamps. To standardize the edge architecture, we extract exactly one continuous primary sensor and the first 24 active binary command streams ($D = 25$) for each channel. We excluded one channel due to unresolvable overlapping ground-truth annotations, yielding 54 evaluated channels. Because timestamps are anonymized, we synthesize communication dropouts by removing 149-step blocks, nominally representing severe multi-minute RF blackouts. This leaves $42,952$ irregular timestamps ($41,298$ valid stride-1 sliding-window positions). We report both strict **Step-Level** metrics and standard **Point-Adjusted (PA)** metrics [4], [11]. No thresholds were tuned on test-set labels.

B. Detection and Precision-Refinement Performance

Table I and Fig. 1 detail end-to-end performance. Layer 1 operates as a sensitive hardware reflex (PA-Recall $= 0.8539$). When its $1,085$ candidate windows are routed to Layer 2, the masked diffusion verifier rejects 574 spurious windows (52.9% false-alarm suppression). Step-Level Precision surges to 0.4524, elevating the overall PA-F1 score from 0.6385 to $\mathbf{0.7004}$. As shown in Fig. 2, Layer 2 markedly improves precision on noisy contextual channels.

C. False-Alarm Suppression and Taxonomy Classification

A major failure mode of edge detectors is flat-channel saturation. Layer 1 routed 83 false-alarm windows on nominal channel `D-5` and 76 on `R-1`. Layer 2's masked diffusion thresholds rejected 100% of false alarms on these quiescent channels. Furthermore, the heuristic physics ratio classifier identifies the anomaly taxonomy (Point vs. Contextual) in 16 of 22 detected channels (72.7%). While this heuristic achieves 72.7% accuracy against SMAP's labels, this is only a marginal improvement over a trivial majority-class baseline ($15/22 = 68.2\%$). It serves as an exploratory heuristic rather than a validated physical diagnostic.

D. Compute Efficiency and Cascade Speedup

In a conventional single-tier deep anomaly detector, a sliding window must be evaluated across all $41,298$ valid stride-1 positions. By contrast, Layer 1 coalesces contiguous burst triggers into just $1,085$ non-overlapping windows. This skips 97.4% of potential diffusion evaluations, yielding a $\mathbf{38.1\times}$ **reduction in potential window evaluations**.

V. Limitations and Discussion

To provide a transparent engineering assessment, we analyze the structural trade-offs of the cascade:

1) **Uncovered Short-Duration Drift:** Layer 1 scores F1 $=$ 0.000 on channels like `E-1` and `G-1`. Because Layer 1 strictly resets at every RF blackout boundary to prevent boundary false alarms, it cannot accumulate long-horizon state memory. Replacing the tree with a Kalman-filtered residual is necessary to capture cross-blackout phase drift.
2) **Sub-Threshold Pruning:** Layer 2 rejected correct candidate windows on low-amplitude channels (`E-4`, `T-1`), reducing overall PA-Recall to 0.7682. The sequence-masked kernel smooths out low-energy oscillations on sparse masks, reflecting the sensitivity–specificity trade-off required for the $+13.37\%$ PA-Precision gain.
3) **High-Activity Command Toggling:** Channels like `B-1` and `A-5` retained partial false-alarm windows because intense scheduled binary command toggling unseen in training inflated auxiliary reconstruction residuals.
4) **Step-Level vs. Point-Adjusted Discrepancy:** In SMAP, ground-truth annotations are broad operational blocks, whereas the physical fault manifests as a localized spike. Because our cascade flags specific timestamps exhibiting reconstruction error rather than padding the window, step-level recall is naturally conservative (35.13%).

VI. Conclusion

We presented a two-tier anomaly triage framework for radar telemetry operating under severe RF blackouts. By pairing a burst-isolated Isolation Forest trigger with an observation-masked frequency-guided diffusion verifier, our system reduces diffusion compute invocations by 97.4% ($38.1\times$ speedup) compared to dense stride-1 baselines. Evaluated on the NASA SMAP benchmark, the cascade achieves a 0.7004 Point-Adjusted F1-score and heuristically classifies Point versus Contextual faults with 72.7% accuracy.

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